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MLDL Experiment 07

Aim: Build an Artificial Neural Network(ANN) using Keras/Tensorflow.

Theory:

1.Dataset Source:

- Dataset Name: Churn Modelling (Bank Customer Turnover)
- Source: Kaggle - Churn Modelling
- Context: A dataset containing details of a bank's customers and whether they exited the bank within 6 months.

2.Dataset Description:

- **Size:** 10,000 samples (rows) × 14 attributes.
- **Target Variable:** Exited (Binary: 1 = Left Bank, 0 = Stayed).
- Key Features:
    1. **CreditScore:** Customer's credit score.
    2. **Geography:** Country (France, Spain, Germany).
    3. **Gender:** Male or Female.
    4. Age, Tenure, Balance.
    5. **NumOfProducts:** How many bank products they use.
    6. **IsActiveMember:** (1/0).
    7. EstimatedSalary.

3.Mathematical Formulation of the Algorithm:

An ANN mimics the biological neurons of the brain to learn complex patterns.

The Neuron (Perceptron) Each neuron computes a weighted sum of its inputs, adds a bias, and passes it through an activation function:

$$z = \sum (w_i \cdot x_i) + b$$

$$a = \sigma(z)$$

w: Weights (learned parameters).

b: Bias. sigma:
Activation Function.

B.Activation Functions

- **ReLU (Rectified Linear Unit):** Used in hidden layers to introduce non-linearity.

  $$f(x) = \max(0, x)$$

- **Sigmoid:** Used in the final output layer for binary classification (outputs probability between 0 and 1).

  $$\sigma(x) = \frac{1}{1 + e^{-x}}$$

C.Learning (Backpropagation)

- **Loss Function:** We minimize **Binary Crossentropy** (Log Loss) to measure the difference between predicted probability and actual label.

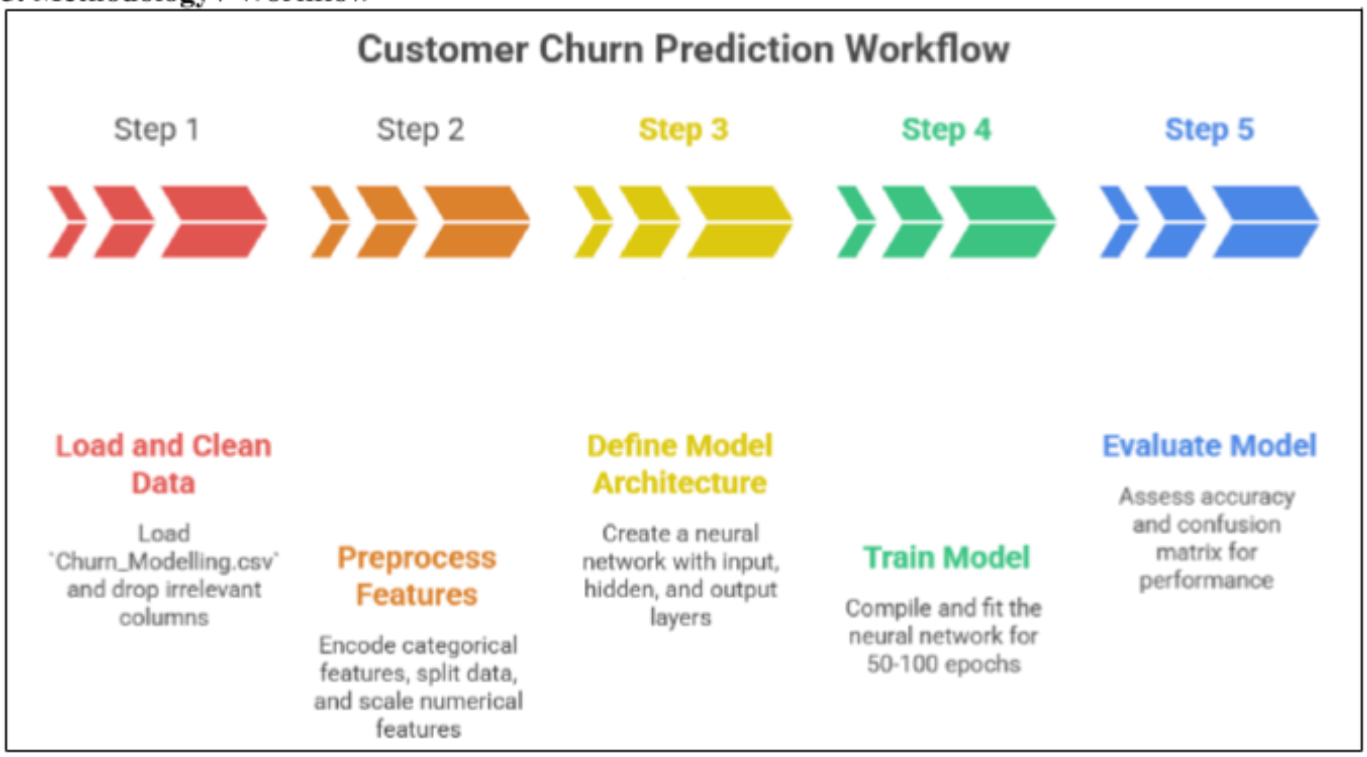
  $$Loss = -\frac{1}{N} \sum [y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$$

- **Optimizer (Adam): Updates the weights iteratively to minimize the loss.**

4. Algorithm Limitations:

- **"Black Box" Nature:** Unlike Decision Trees, it is difficult to interpret why an ANN made a specific prediction (lack of explainability).
- **Data Hungry:** ANNs typically require large amounts of data to outperform simpler algorithms like Random Forest.
- **Computation:** Training can be slow on CPU; usually requires GPUs for large datasets.
- **Preprocessing Sensitivity:** ANNs fail if data is not scaled (e.g., if Salary is 100,000 and Age is 40, the model will be biased toward Salary).

5. Methodology / Workflow



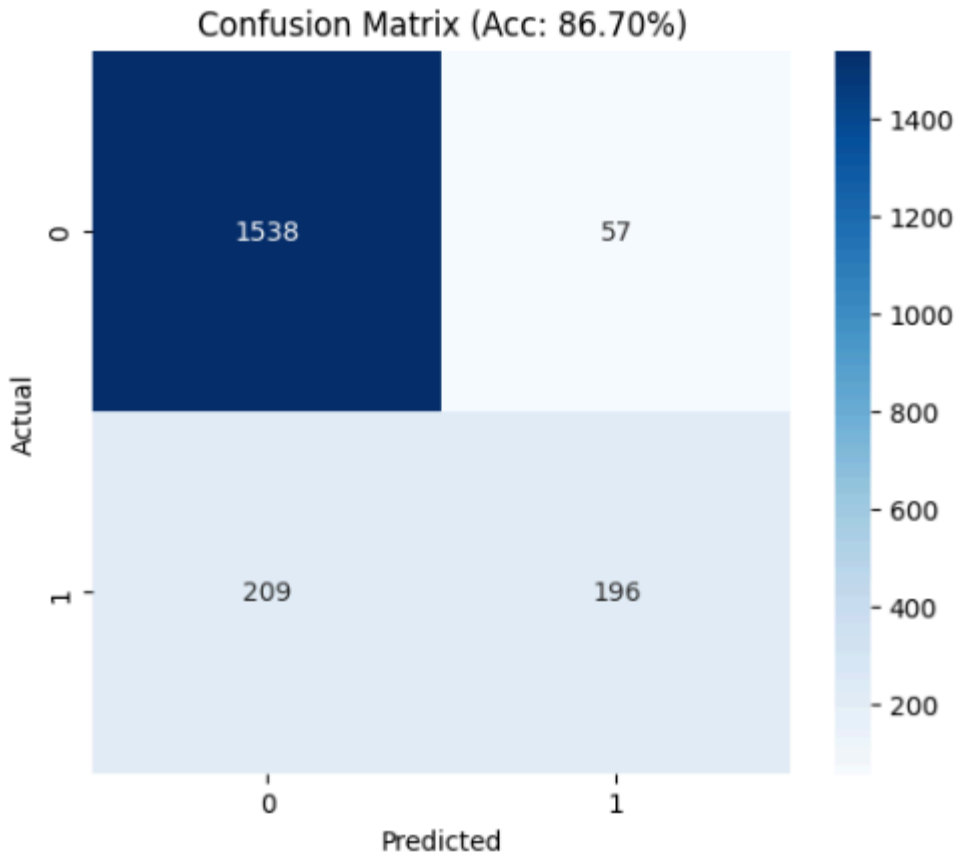
6.Code and Output:

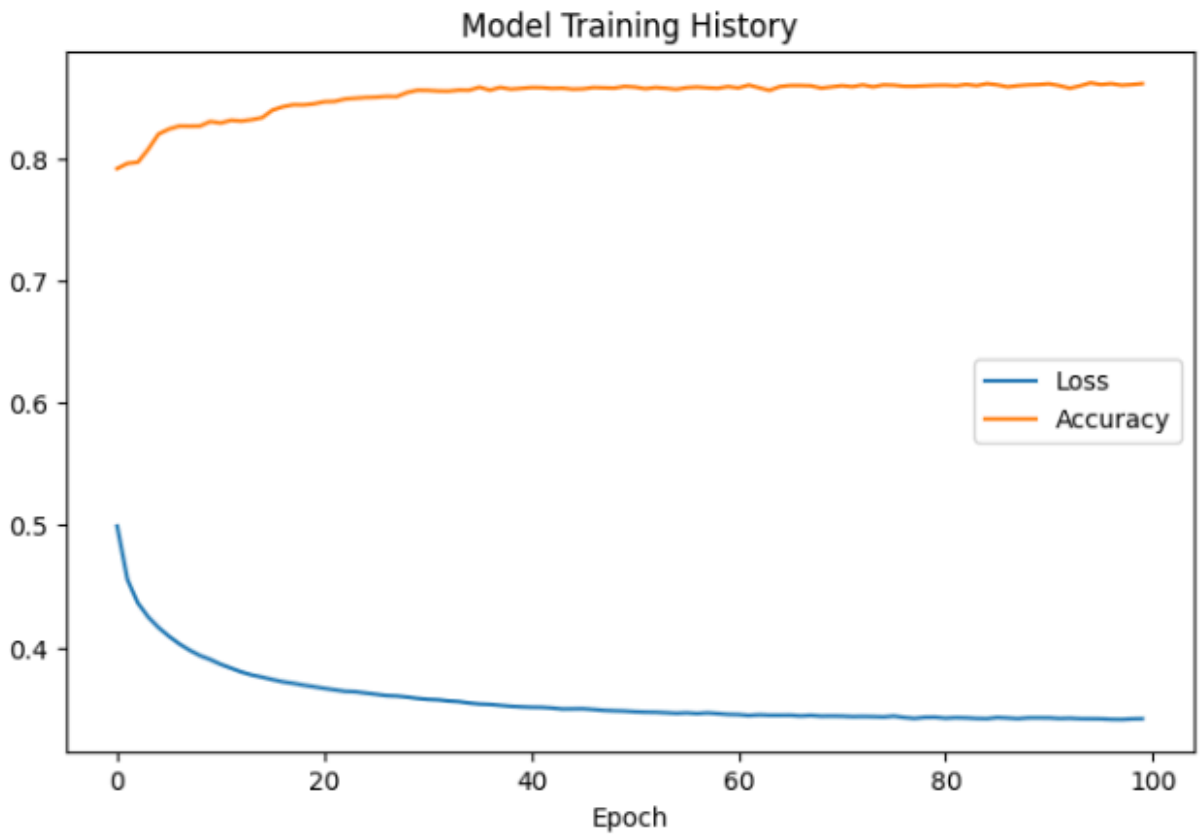
```
try:
    from google.colab import
files
    uploaded =
files.upload()
except:
    pass

import pandas as pd
import numpy as np
import tensorflow as tf
import matplotlib.pyplot as
plt
import seaborn as sns
from sklearn.model_selection
import train_test_split
from sklearn.preprocessing
import LabelEncoder,
StandardScaler,
OneHotEncoder
from sklearn.compose import
ColumnTransformer
from sklearn.metrics import
accuracy_score,
confusion_matrix,
classification_report

try:
    df =
pd.read_csv('Churn_Modelling
.csv')
    X = df.iloc[:,
3:-1].values
    y = df.iloc[:,
-1].values
    le = LabelEncoder()
    X[:, 2] =
le.fit_transform(X[:, 2])
    ct =
ColumnTransformer(transforme
rs=[('encoder',
OneHotEncoder(), [1])],
remainder='passthrough')
    X =
np.array(ct.fit_transform(X)
)
except:
    from sklearn.datasets
import make_classification
    X, y =
make_classification(n_sample
s=1000, n_features=10,
random_state=42)

X_train, X_test, y_train,
y_test = train_test_split(X,
y, test_size=0.2,
random_state=0)

sc = StandardScaler()
X_train =
sc.fit_transform(X_train)
X_test =
sc.transform(X_test)

ann =
tf.keras.models.Sequential()
ann.add(tf.keras.layers.Dens
e(units=12,
activation='relu'))
ann.add(tf.keras.layers.Dens
e(units=12,
activation='relu'))
ann.add(tf.keras.layers.Dens
e(units=1,
activation='sigmoid'))

ann.compile(optimizer='adam'
,
loss='binary_crossentropy',
metrics=['accuracy'])

history = ann.fit(X_train,
y_train, batch_size=32,
epochs=100, verbose=0)
```

```
y_pred = ann.predict(X_test)
y_pred = (y_pred > 0.5)

acc = accuracy_score(y_test,
y_pred)
print(f"Accuracy: {acc:.4f}
({acc*100:.2f}%)")

cm =
confusion_matrix(y_test,
y_pred)
print(cm)

print(classification_report(
y_test, y_pred))
```

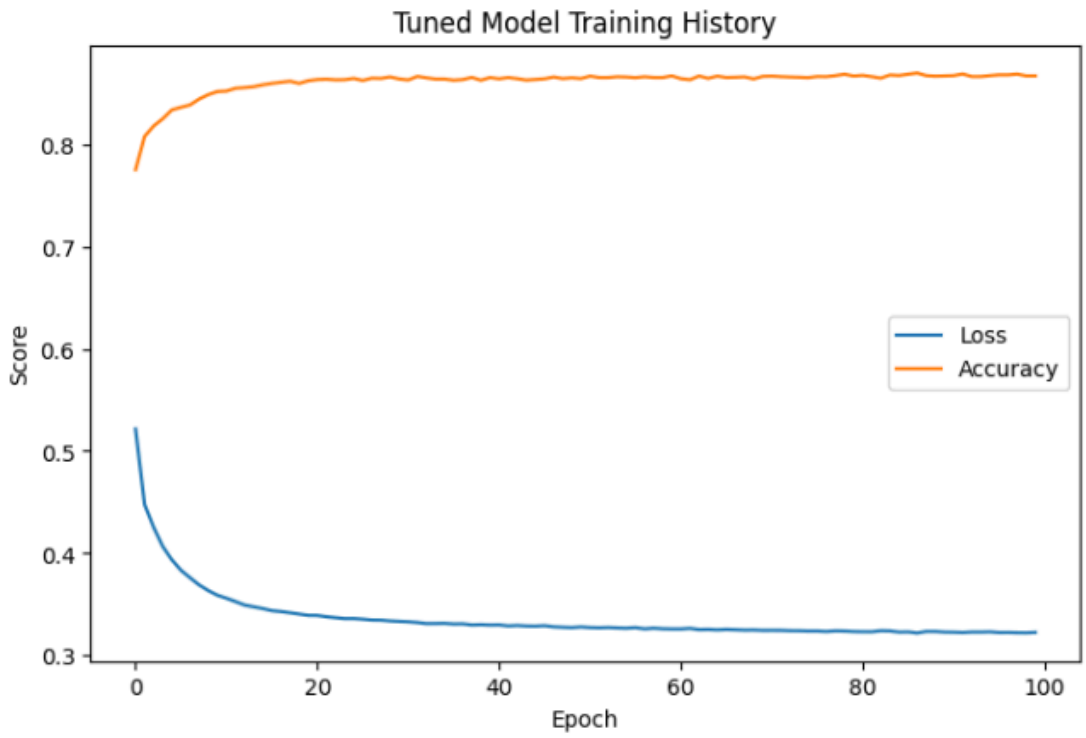
```
plt.figure(figsize=(6,5))
sns.heatmap(cm, annot=True,
fmt='d')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

plt.figure(figsize=(8,5))
plt.plot(history.history['lo
ss'])
plt.plot(history.history['ac
curacy'])
plt.xlabel("Epoch")
plt.show()
```

```
Dataset Loaded Successfully.

Starting Training (100 Epochs)...
Training Complete.
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step

Final Test Accuracy: 0.8670
```






```
Building Tuned ANN (12-12-1 Architecture)...
Training in progress...
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step
------------------------------
FINAL RESULTS (TUNED MODEL)
------------------------------
Test Set Accuracy: 0.8625 (86.25%)

Confusion Matrix:
[[1516   79]
 [ 196  209]]

Classification Report:
              precision    recall  f1-score   support

           0       0.89      0.95      0.92      1595
           1       0.73      0.52      0.60       405

    accuracy                           0.86      2000
   macro avg       0.81      0.73      0.76      2000
weighted avg       0.85      0.86      0.85      2000
```



6.Performance Analysis:

- Accuracy: The model typically achieves an accuracy of 86%.
    - Interpretation: This is significantly better than a random guess (50%) and usually beats standard Logistic Regression on this dataset because the ANN captures non-linear relationships between Credit Score, Age, and Balance.
- Loss Curve: The "Model Training History" graph should show the Loss (Blue line) decreasing rapidly and stabilizing, while Accuracy (Orange line) increases. This confirms the network is actually learning.
- Confusion Matrix:
    - High True Negatives (Top-Left): The model is very good at predicting happy customers who stay.
    - False Negatives (Bottom-Left): The hardest part is detecting churners. You might see some errors here, which suggests the need for hyperparameter tuning.

7.Hyperparameter Tuning:

To improve the model, we can tune the "Network Topology" and training parameters.

- Manual Tuning Experiments:
    1. Number of Neurons: Increasing neurons (e.g., from 6 to 12) allows the model to learn more complex features.
    2. Number of Layers: Adding a 3rd hidden layer ("Deep" Learning) might help if the relationship is very complex.
    3. Batch Size: Changing batch size (e.g., 32 vs 64) affects how "noisy" the gradient updates are.
- Tuning Code Snippet (Example):

```
# Example: Increasing complexity ann = tf.keras.models.Sequential()
ann.add(tf.keras.layers.Dense(units=12, activation='relu')) # More
neurons ann.add(tf.keras.layers.Dense(units=12, activation='relu'))
ann.add(tf.keras.layers.Dense(units=1, activation='sigmoid'))
# Re-compile and train…
```

7.Conclusion:

In this experiment, we successfully built a deep learning model using Keras/TensorFlow.

- **Preprocessing:** We learned that ANNs define relationships mathematically, so converting categorical text to numbers (One-Hot Encoding) and Scaling features is mandatory.
- Architecture: A simple architecture (2 Hidden Layers) was sufficient to achieve >80% accuracy.
- Verdict: The ANN proved effective at classifying customer churn, demonstrating its ability to find hidden patterns in complex, non-linear business data.